

posit_loss_project - [C:/posit_loss_project/vivado/vivado_build/posit_loss_project.xpr] - Vivado 2025.2

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Synthesis Complete ✓

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Report Timing Summary

SYNTHESIZED DESIGN - xc7z030ffg676-1

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report_timing_summary

INFO: [Timing 38-35] Done setting XDC timing constraints.

INFO: [Timing 38-91] UpdateTimingParams: Speed grade: -1, Delay Type: min_max.

INFO: [Timing 38-191] Multithreading enabled for timing update using a maximum of 2 CPUs

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| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14 19:35:11 GMT 2025

| Date : Sat Apr 25 14:21:33 2026

| Host : LAPTOP-SU6IC0U7 running 64-bit major release (build 9200)

| Command : report_timing_summary

| Design : top_wrapper

| Device : 7z030-ffg676

| Speed File : -1 PRODUCTION 1.12 2019-11-22

| Design State : Synthesized

Timing Summary Report

| Timer Settings

| -----

Enable Multi Corner Analysis : Yes

Enable Pessimism Removal : Yes

Pessimism Removal Resolution : Nearest Common Node

Enable Input Delay Default Clock : No

Enable Preset / Clear Arcs : No

Disable Flight Delays : No

Ignore I/O Paths : No

Timing Early Launch at Borrowing Latches : No

Borrow Time for Max Delay Exceptions : Yes

Merge Timing Exceptions : Yes

Type a Tcl command here

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Name	Max Paths	Min Paths
Slow	Yes	Yes
Fast	Yes	Yes

Report Methodology

No report available as report_methodology has not been run prior. Run report_methodology on the current design for the summary of methodology violations.

check_timing report

Table of Contents

- 1. checking no_clock (0)
- 2. checking constant_clock (0)
- 3. checking pulse_width_clock (0)
- 4. checking unconstrained_internal_endpoints (0)
- 5. checking no_input_delay (28)
- 6. checking no_output_delay (36)
- 7. checking multiple_clock (0)
- 8. checking generated_clocks (0)
- 9. checking loops (0)
- 10. checking partial_input_delay (0)
- 11. checking partial_output_delay (0)
- 12. checking latch_loops (0)

- 1. checking no_clock (0)

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```
12. checking latch_loops (0)

1. checking no_clock (0)
-----
There are 0 register/latch pins with no clock.

2. checking constant_clock (0)
-----
There are 0 register/latch pins with constant_clock.

3. checking pulse_width_clock (0)
-----
There are 0 register/latch pins which need pulse_width check

4. checking unconstrained_internal_endpoints (0)
-----
There are 0 pins that are not constrained for maximum delay.

There are 0 pins that are not constrained for maximum delay due to constant clock.

5. checking no_input_delay (28)
-----
There are 28 input ports with no input delay specified. (HIGH)

There are 0 input ports with no input delay but user has a false path constraint.

6. checking no_output_delay (36)
-----
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7. checking multiple_clock (0)

There are 0 register/latch pins with multiple clocks.

8. checking generated_clocks (0)

There are 0 generated clocks that are not connected to a clock source.

9. checking loops (0)

There are 0 combinational loops in the design.

10. checking partial_input_delay (0)

There are 0 input ports with partial input delay specified.

11. checking partial_output_delay (0)

There are 0 ports with partial output delay specified.

12. checking latch_loops (0)

There are 0 combinational latch loops in the design through latch input

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Type a Tcl command here

Design Timing Summary

WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
3.883	0.000	0	788	0.066	0.000	0	788	4.650	0.000	0	414

All user specified timing constraints are met.

Clock Summary

Clock	Waveform(ns)	Period(ns)	Frequency(MHz)
sys_clk	{0.000 5.000}	10.000	100.000

Intra Clock Table

Clock Endpoints	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
sys_clk	3.883	0.000	0	788	0.066	0.000	0	788	4.650	0.000	0	414

Inter Clock Table

From Clock	To Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints
------------	----------	---------	---------	-----------------------	---------------------	---------	---------	-----------------------	---------------------

Other Path Groups Table

Path Group From Clock To Clock WNS(ns) TNS(ns) TNS Failing Endpoints TNS Total Endpoints WHS(ns) THS(ns) THS Failing Endpoints THS Total Endpoints

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Path Group	From Clock	To Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total

| Timing Details

| -----

From Clock: sys_clk
To Clock: sys_clk

Setup : 0 Failing Endpoints, Worst Slack 3.883ns, Total Violation 0.000ns
Hold : 0 Failing Endpoints, Worst Slack 0.066ns, Total Violation 0.000ns
PW : 0 Failing Endpoints, Worst Slack 4.650ns, Total Violation 0.000ns

| Timing Details

From Clock: sys_clk

To Clock: sys_clk

Setup :	0 Failing Endpoints, Worst Slack	3.883ns, Total Violation	0.000ns
Hold :	0 Failing Endpoints, Worst Slack	0.066ns, Total Violation	0.000ns
PW :	0 Failing Endpoints, Worst Slack	4.650ns, Total Violation	0.000ns

Max Delay Paths

Slack (MET) : 3.883ns (required time - arrival time)

Source: m1/u_enc/idx_reg_rep[4]/C
(rising edge-triggered cell FDCE clocked by sys_clk {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: m1/u_enc/best_code_reg[0]/CE
(rising edge-triggered cell FDCE clocked by sys_clk {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: sys_clk

Path Type: Setup (Max at Slow Process Corner)

Requirement: 10.000ns (sys_clk rise@10.000ns - sys_clk rise@0.000ns)

Data Path Delay: 5.696ns (logic 1.827ns (32.075%) route 3.869ns (67.925%))

Logic Levels: 10 (CARRY4=4 LUT1=1 LUT3=1 LUT4=1 LUT6=3)

Clock Path Skew: -0.145ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 1.673ns = (11.673 - 10.000)

Source Clock Delay (SCD): 1.977ns
 Clock Pessimism Removal (CPR): 0.159ns
 Clock Uncertainty: 0.035ns $((TSJ^2 + TIJ^2)^{1/2} + DJ) / 2 + PE$
 Total System Jitter (TSJ): 0.071ns
 Total Input Jitter (TIJ): 0.000ns
 Discrete Jitter (DJ): 0.000ns
 Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

(clock sys_clk rise edge)		0.000	0.000	r
		0.000	0.000	r clk (IN)
net (fo=0)		0.000	0.000	clk
IBUF (Prop_ibuf_I_O)		0.743	0.743	r clk_IBUF_inst/O
net (fo=1, unplaced)		0.530	1.273	clk_IBUF
BUFG (Prop_bufg_I_O)		0.120	1.393	r clk_IBUF_BUFG_inst/O
net (fo=413, unplaced)		0.584	1.977	m1/u_enc/clk_IBUF_BUFG
FDCE				r m1/u_enc/idx_reg_rep[4]/C

FDCE (Prop_fdce_C_Q)		0.281	2.258	f m1/u_enc/idx_reg_rep[4]/Q
net (fo=35, unplaced)		0.731	2.989	m1/u_enc/idx[4]
LUT6 (Prop_lut6_I0_O)		0.153	3.142	r m1/u_enc/delta_carry_i_11/O
net (fo=1, unplaced)		0.521	3.663	m1/u_enc/delta_carry_i_11_n_0
LUT6 (Prop_lut6_I1_O)		0.053	3.716	r m1/u_enc/delta_carry_i_4/O
net (fo=1, unplaced)		0.000	3.716	m1/u_enc/delta_carry_i_4_n_0
CARRY4 (Prop_carry4_S[0]_O[0])				
		0.143	3.859	f m1/u_enc/delta_carry/O[0]
net (fo=4, unplaced)		0.266	4.125	m1/u_enc/delta[0]
LUT1 (Prop_lut1_I0_O)		0.153	4.278	r m1/u_enc/best_diff[4]_i_3/O
net (fo=1, unplaced)		0.247	4.525	m1/u_enc/p_0_in[0]
CARRY4 (Prop_carry4_CYINIT_O[1])				

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0.416 4.941 f m1/u_enc/best_diff_reg[4]_i_2/O[1]
net (fo=2, unplaced) 0.463 5.404 m1/u_enc/abs_delta0[2]
LUT3 (Prop_lut3_I_O) 0.155 5.559 f m1/u_enc/best_code0_carry_i_11/O
net (fo=2, unplaced) 0.351 5.910 m1/u_enc/in8[2]
LUT6 (Prop_lut6_I_O) 0.053 5.963 r m1/u_enc/best_code0_carry_i_3/O
net (fo=1, unplaced) 0.363 6.326 m1/u_enc/best_code0_carry_i_3_n_0
CARRY4 (Prop_carry4_DI[1]_CO[3])
0.307 6.633 r m1/u_enc/best_code0_carry/CO[3]
net (fo=1, unplaced) 0.000 6.633 m1/u_enc/best_code0_carry_n_0
CARRY4 (Prop_carry4_CI_CO[3])
0.060 6.693 r m1/u_enc/best_code0_carry__0/CO[3]
net (fo=1, unplaced) 0.518 7.211 m1/u_enc/best_code0_carry__0_n_0
LUT4 (Prop_lut4_I_O) 0.053 7.264 r m1/u_enc/best_diff[15]_i_1/O
net (fo=24, unplaced) 0.409 7.673 m1/u_enc/best_diff[15]_i_1_n_0
FDCE r m1/u_enc/best_code_reg[0]/CE

```

```

(clock sys_clk rise edge) 10.000 10.000 r
0.000 10.000 r clk (IN)
net (fo=0) 0.000 10.000 clk
IBUF (Prop_ibuf_I_O) 0.617 10.617 r clk_IBUF_inst/O
net (fo=1, unplaced) 0.503 11.121 clk_IBUF
BUFG (Prop_bufg_I_O) 0.113 11.234 r clk_IBUF_BUFG_inst/O
net (fo=413, unplaced) 0.439 11.673 m1/u_enc/clk_IBUF_BUFG
FDCE r m1/u_enc/best_code_reg[0]/C
clock pessimism 0.159 11.832
clock uncertainty -0.035 11.796
FDCE (Setup_fdce_C_CE) -0.241 11.555 m1/u_enc/best_code_reg[0]

```

```

required time 11.555
arrival time -7.673

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slack

3.883

Min Delay Paths

Slack (MET) : 0.066ns (arrival time - required time)
Source: m1/enc_start_reg/C
(rising edge-triggered cell FDCE clocked by sys_clk {rise@0.000ns fall@5.000ns period=10.000ns})
Destination: m1/enc_start_reg/D
(rising edge-triggered cell FDCE clocked by sys_clk {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: sys_clk
Path Type: Hold (Min at Fast Process Corner)
Requirement: 0.000ns (sys_clk rise@0.000ns - sys_clk rise@0.000ns)
Data Path Delay: 0.280ns (logic 0.168ns (59.973%) route 0.112ns (40.027%))
Logic Levels: 1 (LUT4=1)
Clock Path Skew: 0.145ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): 0.834ns
Source Clock Delay (SCD): 0.507ns
Clock Pessimism Removal (CPR): 0.182ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
(clock sys_clk rise edge)		0.000	0.000	r
		0.000	0.000	r clk (IN)
net (fo=0)		0.000	0.000	clk
IBUF (Prop_ibuf_I_O)		0.143	0.143	r clk_IBUF_inst/O
net (fo=1, unplaced)		0.223	0.367	clk_IBUF

BUFG (Prop_bufg_I_O)	0.026	0.393	r	clk_IBUF_BUFG_inst/O
net (fo=413, unplaced)	0.114	0.507	m1/clk_IBUF_BUFG	
FDCE			r	m1/enc_start_reg/C

FDCE (Prop_fdce_C_Q)	0.104	0.611	r	m1/enc_start_reg/Q
net (fo=4, unplaced)	0.112	0.723	m1/enc_start_reg_n_0	
LUT4 (Prop_lut4_I3_O)	0.064	0.787	r	m1/enc_start_i_1/O
net (fo=1, unplaced)	0.000	0.787	m1/enc_start_i_1_n_0	
FDCE			r	m1/enc_start_reg/D

(clock sys_clk rise edge)	0.000	0.000	r	
	0.000	0.000	r	clk (IN)
net (fo=0)	0.000	0.000	clk	
IBUF (Prop_ibuf_I_O)	0.310	0.310	r	clk_IBUF_inst/O
net (fo=1, unplaced)	0.235	0.545	clk_IBUF	
BUFG (Prop_bufg_I_O)	0.030	0.575	r	clk_IBUF_BUFG_inst/O
net (fo=413, unplaced)	0.259	0.834	m1/clk_IBUF_BUFG	
FDCE			r	m1/enc_start_reg/C
clock pessimism	-0.182	0.652		
FDCE (Hold_fdce_C_D)	0.069	0.721	m1/enc_start_reg	

required time	-0.721
arrival time	0.787

slack	0.066
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Pulse Width Checks

Clock Name: sys_clk
 Waveform(ns): { 0.000 5.000 }
 Period(ns): 10.000
 Sources: { clk }

Check Type	Corner	Lib Pin	Reference Pin	Required(ns)	Actual(ns)	Slack(ns)	Location	Pin
Min Period	n/a	DSP48E1/CLK	n/a	1.825	10.000	8.175	m1/prod_q0/CLK	
Low Pulse Width	Slow	FDCE/C	n/a	0.350	5.000	4.650	loss_q_reg[15]_i_2/C	
High Pulse Width	Slow	FDCE/C	n/a	0.350	5.000	4.650	loss_q_reg[15]_i_2/C	

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Design Timing Summary

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Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.883 ns	Worst Hold Slack (WHS): 0.066 ns	Worst Pulse Width Slack (WPWS): 4.650 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 788	Total Number of Endpoints: 788	Total Number of Endpoints: 414

All user specified timing constraints are met.

Timing Summary - timing_1

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Settings

Enable Multi Corner Analysis:Yes

Enable Pessimism Removal:Yes

Pessimism Removal Resolution:Nearest Common Node

Enable Input Delay Default Clock:No

Enable Preset / Clear Arcs:No

Disable Flight Delays:No

Ignore I/O Paths:No

Timing Early Launch at Borrowing Latches:No

Borrow Time for Max Delay Exceptions:Yes

Merge Timing Exceptions:Yes

Inter-SLR Compensation:Conservative

Multi-Corner Configuration

Corner Name	Analyze Max Paths	Analyze Min Paths
Slow	Yes	Yes
Fast	Yes	Yes

Timing Summary - timing_1

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Clock Summary

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Name	Waveform	Period (ns)	Frequency (MHz)
sys_clk	{0.000 5.000}	10.000	100.000

Timing Summary - timing_1

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Clock	Edges (WNS)	WNS (ns)	TNS (ns)	Failing Endpoints (TNS)	Total Endpoints (TNS)	Edges (WHS)	WHS (ns)	THS (ns)	Failing Endpoints (THS)	Total Endpoints (THS)	WPWS (ns)	TPWS (ns)	Failing Endpoints (TPWS)
sys_clk	rise - rise	3.883	0.000	0	788	rise - rise	0.066	0.000	0	788	4.650	0.000	0

Timing Summary - timing_1